光之殇

Lost in the Glare



昆虫绕着灯飞，真只是因为喜欢光吗？

英文原文

A moth **skims over** an ultraviolet lamp and flips upside down, **belly** to the sky. In Costa Rica's **misty** Monteverde **cloud-forest**, **stereo** cameras caught such **acrobatics** for a 2024 Imperial College study that **unravelled** an **age-old** puzzle, the **helpless swirl** of insects round **porch** lights.

Folk wisdom blames the circling on **attraction**. In fact, insects **instinctively** keep their backs to whatever shines brightest, normally the sky, to stay level. A lamp, a **counterfeit** sky, **hijacks** this **built-in** habit, leaving insects **eddying** in its **halo** or **inverted** above it. In 477 field **clips**, **precious few darted** straight at the light.

Many fireflies **flirt** with their **lanterns**. A male **blinks** an **overture**, a female **winks** back, and bright nights **smother** the **banter**. At Tufts University, **suitors** under artificial light **mustered** about five flash patterns a minute instead of ten. Females **reciprocated** about half as often under red or blue light, scarcely at all under white, and never under **amber**.

The harm **spills into** the **ecosystem**. Along 55 Thames Valley **roadsides**, **lamp-lit hedges** held 47% fewer moth **caterpillars** than dark ones, with **harsh** white LEDs worse than **mellower sodium lamps**. Street lamps **rigged up** in seven of 14 Swiss **meadows** cut **nocturnal** pollinator visits by 62%. The plants set fewer fruit, and daytime pollinators proved poor **understudies**.

Over 80% of **humanity** lives beneath a **luminous veil**, which hides the pale **ribbon** of the Milky Way from more than a third. **Citizen** scientists' star counts from 2011 to 2022 show the sky brightening about 9.6% a year. At that **pace**, a child born where 250 stars **twinkle** would see about 100 by their eighteenth birthday.

中文译文

一只飞蛾掠过一盏紫外灯，随即上下颠倒地翻过身来，肚皮朝天。在哥斯达黎加蒙特韦尔德雾气缭绕的云雾林里，立体摄像机为伦敦帝国理工学院2024年的一项研究拍下了这类杂技动作。这项研究解开了一个由来已久的谜团，那就是昆虫为什么会身不由己地绕着门廊灯打转。

民间的老说法把这种打转归咎于吸引。事实上，昆虫会本能地让背部朝向最亮的东西，那通常是天空，以此保持平稳。灯成了一片冒牌的天空，劫持了这种与生俱来的习性，让昆虫在它的光晕里打着旋，或在它上方翻转过来。在477段野外视频片段中，径直冲向灯光的昆虫寥寥无几。

许多萤火虫用自己的灯笼来调情。雄虫闪烁着奏出一段序曲，雌虫眨眼般回应，而太亮的夜晚会淹没这番调笑。在塔夫茨大学，处在人造光下的求偶者每分钟只勉强打出约五组闪光，而不是十组。雌虫在红光或蓝光下回应的次数约减半，在白光下几乎不回应，在琥珀色光下则从不回应。

危害还会蔓延到整个生态系统。在泰晤士河谷的55处路边，有灯照明的树篱里，蛾类毛虫比没灯的地方少47%，而刺眼的白色LED比更柔和的钠灯更糟。在瑞士14片草甸中的7片，临时架起的路灯让夜间传粉者的造访减少了62%。植物结的果实变少了，而白天的传粉者也只是不称职的替补。

全球80%以上的人类生活在一层发光的帷幕之下，它让三分之一以上的人看不见银河那条淡淡的光带。公民科学家2011年至2022年的数星记录显示，夜空每年约变亮9.6%。按这个速度，若一个孩子出生在能看到250颗星星闪耀的地方，到18岁生日时只能看到约100颗。

词汇与表达

skims over v.
掠过
A moth skims over an ultraviolet lamp

belly n.
肚皮

misty adj.
雾气缭绕的

cloud-forest n.
云雾林

stereo adj.
立体

acrobatics n.
杂技

unravelled v.
解开了
unravelled an age-old puzzle

age-old adj.
由来已久的
an age-old puzzle

helpless adj.
身不由己地

swirl n.
打转

porch n.
门廊

Folk wisdom n.
民间的老说法
Folk wisdom blames the circling

attraction n.
吸引
blames the circling on attraction

instinctively adv.
本能地
insects instinctively keep their backs

counterfeit adj.
冒牌的
a counterfeit sky

hijacks v.
劫持了
hijacks this built-in habit

built-in adj.
与生俱来的
this built-in habit

eddying v.
打着旋

halo n.
光晕

inverted adj.
翻转
inverted above it

clips n.
视频片段

precious few phr.
寥寥无几
precious few darted straight

darted v.
径直冲

flirt v.
调情

lanterns n.
灯笼

blinks v.
闪烁

overture n.
序曲

词汇与表达（续）

winks v.
眨眼

smother v.
淹没

banter n.
调笑

suitors n.
求偶者

mustered v.
勉强打出

reciprocated v.
回应
Females reciprocated

amber n.
琥珀色

spills into v.
蔓延到

ecosystem n.
生态系统
The harm spills into the ecosystem

roadsides n.
路边

lamp-lit adj.
有灯照明的

hedges n.
树篱

caterpillars n.
毛虫
47% fewer moth caterpillars

harsh adj.
刺眼的
harsh white LEDs

mellower adj.
更柔和的

sodium lamps n.
钠灯
mellower sodium lamps

rigged up v.
临时架起
Street lamps rigged up

meadows n.
草甸

nocturnal adj.
夜间
nocturnal pollinator visits

understudies n.
替补
proved poor understudies

humanity n.
人类

luminous adj.
发光的
a luminous veil

veil n.
帷幕

ribbon n.
光带

Citizen n.
公民
Citizen scientists' star counts

pace n.
速度
At that pace

twinkle v.
闪耀

文章简述

- 飞行：昆虫本来靠把背部朝向明亮天空保持平衡。2024年研究发现，路灯成了“假天空”，昆虫把背转向灯，反而绕圈甚至翻身。
- 求偶：萤火虫靠闪光寻找配偶。实验中，人造光使雄虫发光减少，雌虫也更少回应，彼此传递的求偶信号受到干扰。
- 幼虫：英国路边调查发现，有灯树篱的蛾类毛虫比暗处少47%；白色LED的影响比钠灯更重。
- 传粉：瑞士实验中，有路灯的草甸比无灯草甸少了62%的夜间传粉昆虫来访，植物结出的果实也更少；白天的昆虫没能补上这份损失。
- 夜空：全球超过八成人口生活在光污染下，超过三分之一看不到银河。灯光不只遮住星星，也在改变昆虫的飞行、繁殖和植物的传粉。